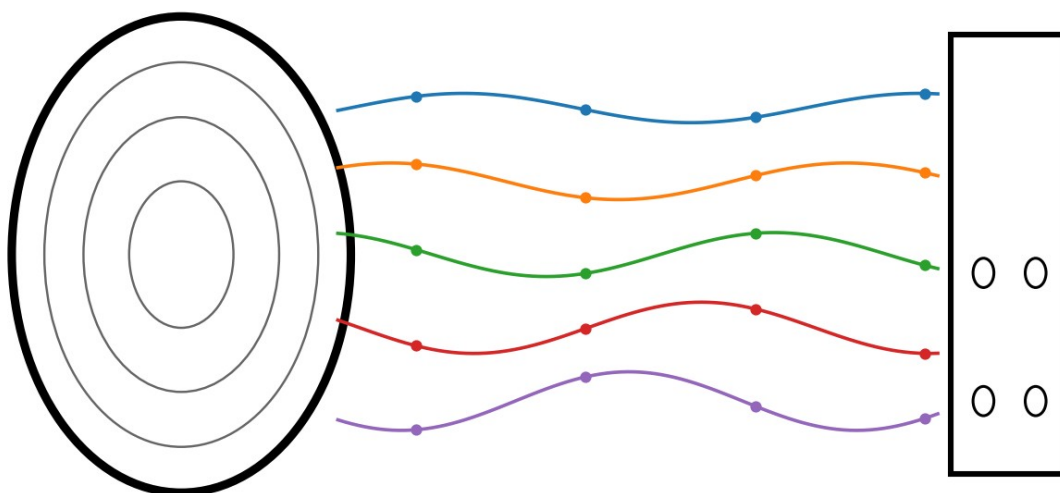




LSC

SOLAR NEUTRINOS, THE GALLIUM ANOMALY AND TESTABLE RESIDUAL PHYSICS

Rebuilt research monograph after the audit of the “27%” claim

Concept author of LSC: Łukasz Żuchowski

English edition • 19 September 2026

Data → PMNS/MSW → residual → LSC → falsification

Note on scientific status

This publication is a research monograph and a reconstruction document for the LSC hypothesis, not a peer-reviewed article or an established physical theory.

In this book, LSC is treated as a hypothesis or effective model that must compete with the standard description of neutrinos: the Standard Solar Model (SSM), PMNS mixing and the MSW effect. Experimental facts, standard interpretation and LSC-specific assumptions are separated on purpose.

The most important correction with respect to the previous edition is the abandonment of the “single neutrino” narrative as the main subject. A single neutrino can illustrate dynamics; it cannot by itself prove a statistical population deficit.

Result of the 27% audit

No canonical frozen statement was found in the checked LSC materials asserting that modern solar data show a universal 27% neutrino loss. The LSC 6.0/6.3 materials instead anchored the gallium anomaly in BEST at $R \approx 0.79$, i.e. about 21%. There is, however, a real literature value extremely close to 27%: for the Haxton 71Ga cross-section model, the global ratio in source-based gallium experiments was $R = 0.731$, i.e. $1 - R = 26.9\%$. This is model-dependent and belongs to the gallium anomaly with 51Cr/37Ar sources, not to a universal solar-neutrino deficit.

Correct verification order for LSC



A mechanism cannot come before establishing whether the data contain a residual at all.

Fig. 1. The rebuilt logic of the book: from data to mechanism, not from mechanism to a pre-decided outcome.

Executive summary

This edition answers the question that should have stood at the beginning of the entire project: what exactly is the “deficit” that LSC is supposed to explain? The answer is not a single number. Historically, the solar-neutrino problem referred to a deficit of electron neutrinos relative to then-current solar models without oscillations. Contemporary physics interprets the bulk of that phenomenon as flavor conversion $\nu_e \rightarrow \nu_\mu/\nu_\tau$, with the decisive evidence coming from SNO.

Therefore one must distinguish among the total active flux, the ν_e fraction, the event count in a specific detector, and the adopted interaction cross section.

A second, separate problem is the gallium anomaly. GALLEX and SAGE, and later BEST, used intense artificial electron-neutrino sources to probe $71\text{Ga}(\nu_e, e^-)71\text{Ge}$. BEST obtained $R \approx 0.79$ in the inner volume and $R \approx 0.77$ in the outer volume. This 20–24% deficit is not a measurement of the Sun. A value close to 27% emerges only under specific cross-section assumptions; in 2026 new calculations again shifted the estimated anomaly size, and an accepted PRL even suggests that a revision of the cross-section theory may largely remove the anomaly.

Consequently, LSC should not be built so as to “hit 27%”. It should define an additional operator or response function, reproduce the standard null model first, and only then be fitted simultaneously to solar and source-based data. The model is scientifically valuable only if one parameter set improves the description of many independent datasets while leaving intact the channels that the standard description already explains well.

Five most important conclusions

1. 27% is not an established contemporary value of a universal solar-neutrino deficit.
2. The closest precise literature trace of 27% is 26.9% in the gallium anomaly for the Haxton cross-section model ($R=0.731$), not a universal solar loss.
3. The LSC 6.0/6.3 materials themselves are anchored much closer to BEST $R \approx 0.79$, i.e. about 21%, which confirms that 27% was not a stable project parameter.
4. SNO showed that a disappearance of ν_e does not mean a disappearance of the total active flux; standard PMNS/MSW must be the baseline.
5. A scientifically defensible LSC must be a residual model: $M1 = \text{SSM} + \text{PMNS/MSW} + \text{LSC}$, statistically compared to $M0 = \text{SSM} + \text{PMNS/MSW}$.

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PART I

WHAT THE EXPERIMENTS ACTUALLY MEASURED

From the historical solar-neutrino problem to the precision era

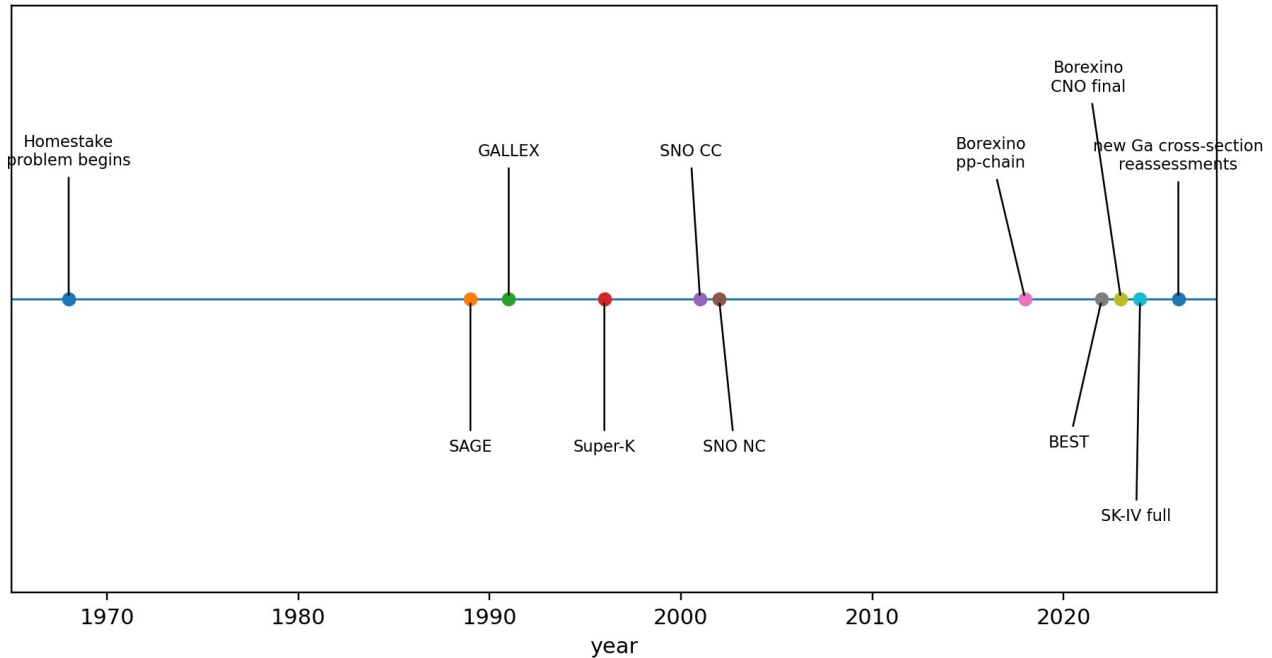


Fig. 2. The main steps from the historical solar-neutrino problem to precision spectroscopy and the modern audit of the gallium anomaly.

1. Defining a deficit — without a denominator there is no percentage

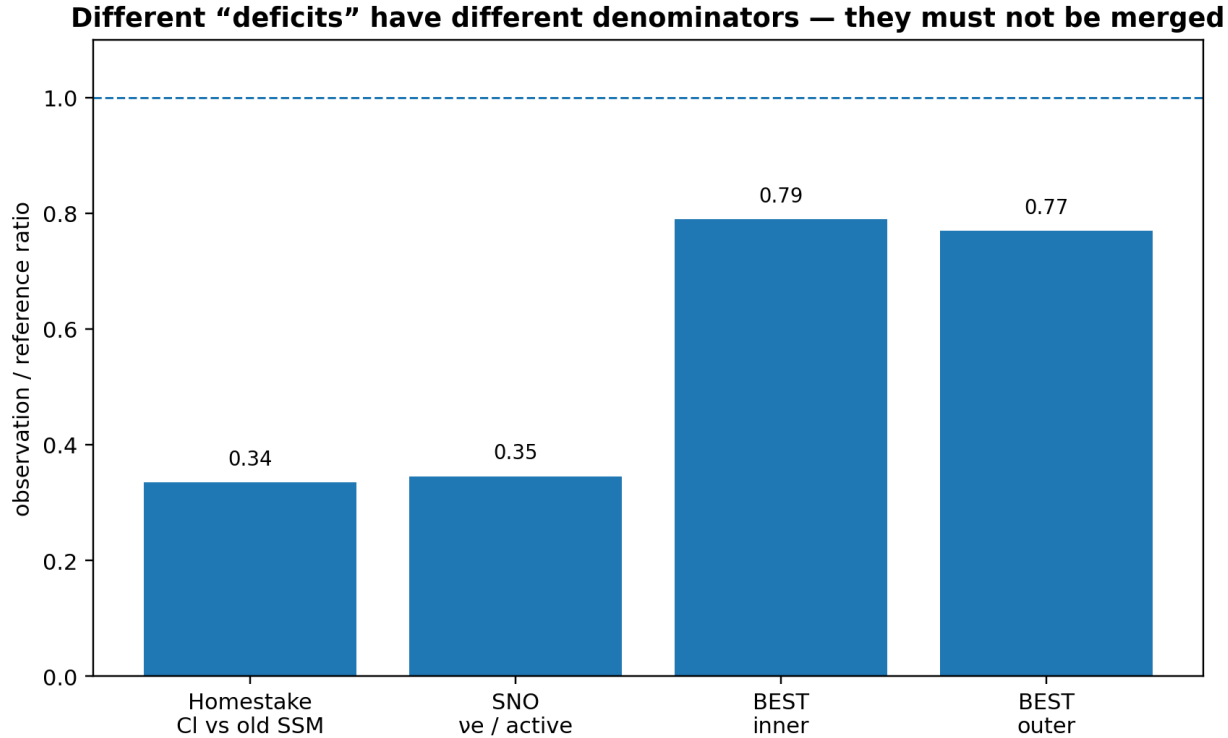
The simplest mistake in this field is to quote a “deficit of X%” without telling the reader what the reference quantity actually is. In solar-neutrino physics, at least four different ratios appear: an observed event rate divided by an old no-oscillation solar-model prediction, a measured electron-neutrino flux divided by the total active flux, a measured rate divided by a PMNS/MSW expectation, and a source-calibration rate divided by a theoretical capture cross section. Those ratios are not interchangeable.

$$D = 1 - N_{\text{obs}} / N_{\text{ref}}$$

In the equation above, D becomes meaningful only after N_{ref} is defined. If N_{ref} is an outdated no-oscillation SSM, the resulting deficit describes a historical anomaly. If N_{ref} is a modern PMNS/MSW calculation, the same percentage becomes a test of residual new physics. If N_{ref} depends on a debated ^{71}Ga cross section, the number partly reflects nuclear theory rather than a neutrino propagation effect.

Symbol	Meaning	Typical uncertainty source
N_{obs}	measured event rate or flux	statistics, detector calibration
N_{ref}	reference prediction	solar model, oscillation parameters, cross section
$R = N_{\text{obs}}/N_{\text{ref}}$	observation-to-reference ratio	combined propagated error
$D = 1 - R$	apparent deficit	depends entirely on the chosen

		denominator
$P_{ee}(E)$	electron-neutrino survival probability	mixing angles, density profile
Φ_{total}	total active flux	channel separation, cross-calibration



These bars are intentionally juxtaposed as an example of non-equivalent definitions, not as a single common measurement.

Fig. 3. “A deficit” is not one thing. Different experiments and channels compare different observables to different baselines.

2. The Sun as a neutrino source

The Sun produces neutrinos in several branches of nuclear burning: the pp chain (pp, pep, 7Be, 8B, hep) and the CNO cycle. Their energies and relative production rates differ strongly. Because matter effects in the Sun depend on density and because oscillation probabilities depend on energy, no single fixed percentage can summarize the full solar-neutrino phenomenology.

- pp neutrinos dominate in number and probe the low-energy, mostly vacuum-oscillation regime.
- 7Be and pep neutrinos probe the transition region.
- 8B neutrinos probe the matter-enhanced MSW regime and dominate the classic water-Cherenkov signal.
- CNO neutrinos are important for solar metallicity tests and the internal composition problem.

Any serious LSC comparison must therefore either fit the energy dependence explicitly or justify a controlled energy-averaged approximation. A single “27% deficit” is at best a shorthand for one limited projection of a multi-dimensional problem.

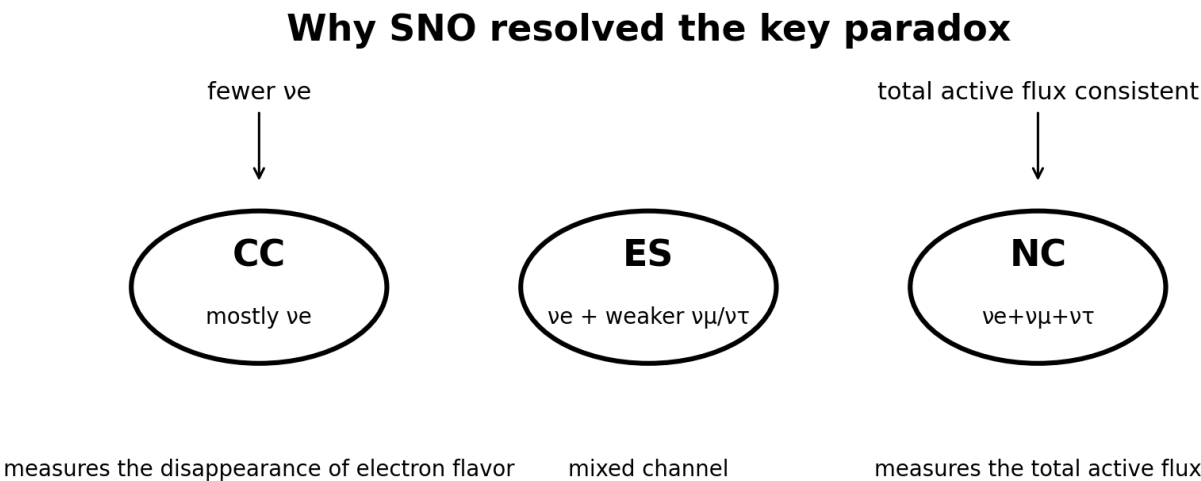
3. The historical solar-neutrino problem

The historical solar-neutrino problem emerged because radiochemical and water-Cherenkov experiments measured fewer electron-neutrino events than expected from standard solar models without oscillations. Homestake, GALLEX, GNO, SAGE and Kamiokande/early Super-K all contributed to this picture. However, those detectors had different thresholds and flavor sensitivities. The “missing neutrinos” were not directly the same quantity in every instrument.

Experiment / channel	Illustrative result	What it really tested
Homestake (Cl)	~2.56 SNU vs ~7.64 SNU	high-energy solar ν_e against an older SSM baseline
GALLEX/SAGE solar Ga	~66 SNU vs older larger estimates	integrated low-energy ν_e capture with strong cross-section dependence
Kamiokande / Super-K ES	suppressed ν_e -sensitive elastic scattering rate	mostly 8B ν_e with some ν_μ/ν_τ sensitivity

Historically, this was enough to motivate neutrino oscillations. But the historical deficit is not automatically a residual anomaly today. That distinction is central to the rebuilt book.

4. SNO and the separation of flavor from total active flux



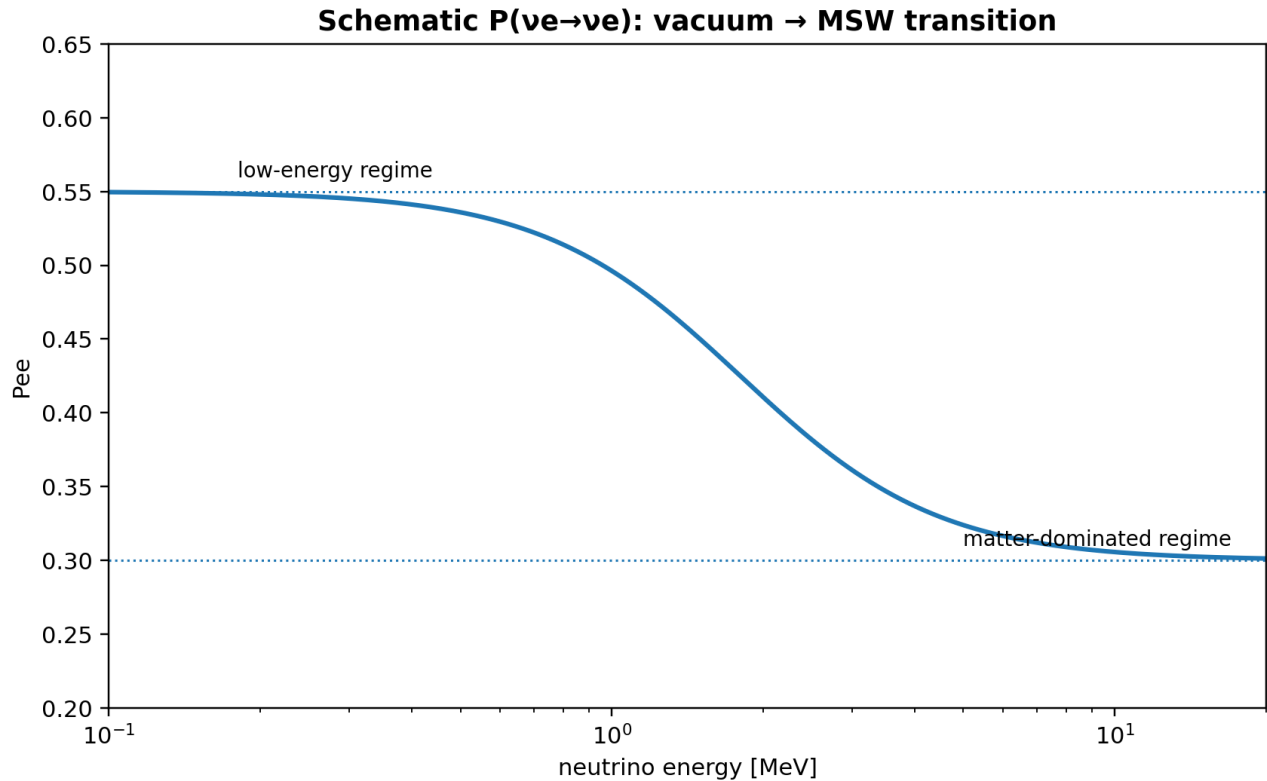
Conclusion: “missing ν_e ” $\neq$ “missing all neutrinos”. Oscillations change flavor.

Fig. 4. SNO separated the disappearance of electron flavor from the total active neutrino flux.

The Sudbury Neutrino Observatory changed the field because it could compare charged-current, elastic-scattering and neutral-current channels in the same program. Charged current is mostly sensitive to ν_e , while neutral current measures the total active flux $\nu_e + \nu_\mu + \nu_\tau$. SNO found that the ν_e component was suppressed, but the total active 8B flux remained compatible with solar-model expectations. This showed that the historical “missing neutrinos” were largely a flavor-conversion problem, not a destruction of the total active neutrino population.

For LSC this is a non-negotiable boundary condition. Any claim of a genuine active-flux loss must either survive the SNO NC constraint or explain precisely why the comparison is not applicable.

5. Borexino, Super-K and the modern baseline



This is a schematic curve, not a fit to data. Its role is to show why there is no single universal “deficit percentage”.

Fig. 5. Schematic electron-neutrino survival probability $P_{ee}(E)$.

Borexino mapped the low-energy solar neutrino flux with unprecedented purity, including pp, ^7Be , pep and later CNO neutrinos. Super-K measured the high-energy spectrum and day-night effect with great precision. Together with global PMNS fits such as NuFIT, they define the baseline any additional model must confront.

The qualitative lesson is simple: $P_{ee}(E)$ is energy dependent. At low energy it sits near the vacuum-averaged regime, and at higher energy it approaches the MSW matter-dominated regime. Therefore a model that predicts one universal deficit value independent of energy is immediately suspicious unless it refers to a very specific averaged observable.

PART II

THE 27% INVESTIGATION

Two problems that the older narrative kept mixing

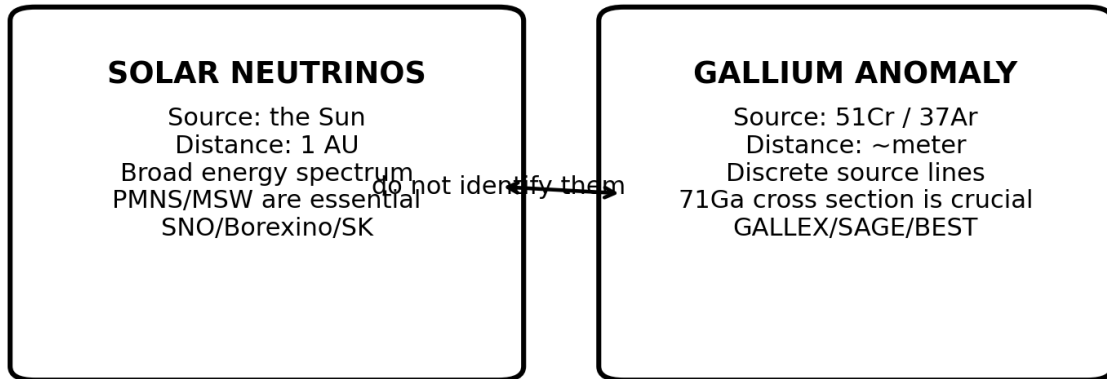


Fig. 6. Solar neutrinos and the gallium anomaly are related historically, but they are not the same problem.

6. BEST and the gallium anomaly

BEST: about 21-23% below expectation for a chosen cross-section model

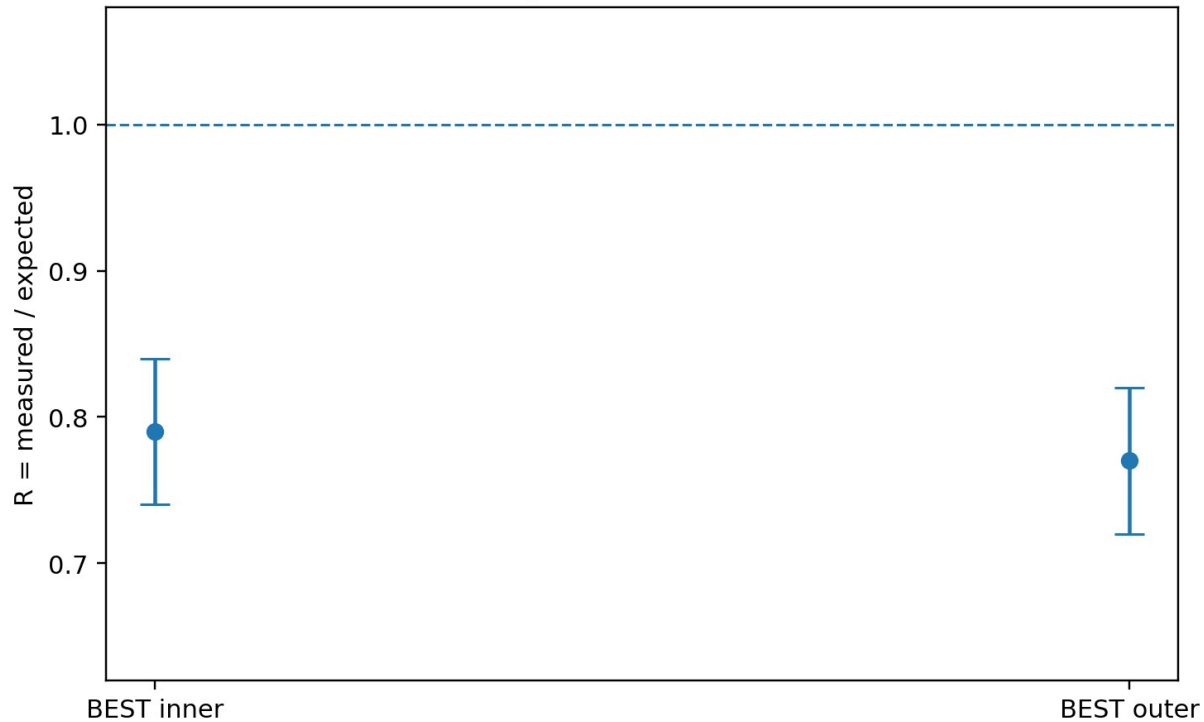


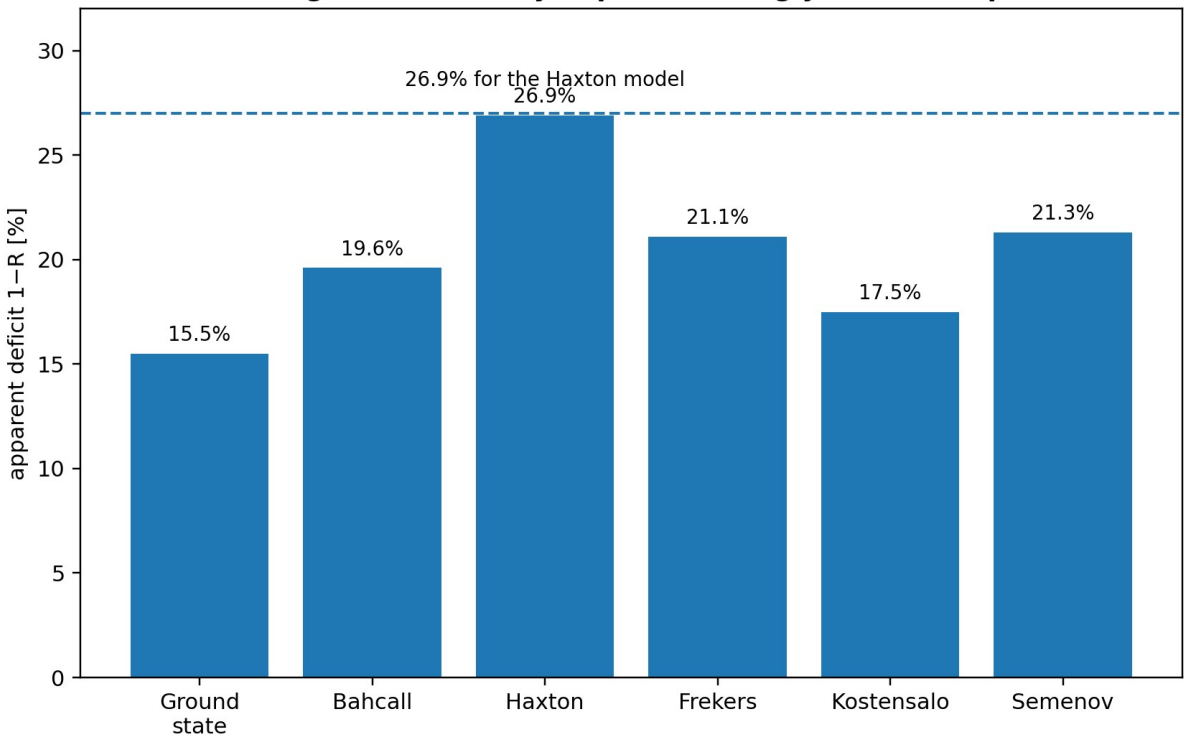
Fig. 7. BEST source-calibration result expressed as $R = \text{measured} / \text{expected}$.

BEST revisited the long-standing gallium calibration problem using an intense ^{51}Cr source and two nested gallium volumes. Its reported ratios are about $R_{\text{inner}} \approx 0.79$ and $R_{\text{outer}} \approx 0.77$, corresponding to apparent deficits of roughly 21% and 23%. These are statistically striking, but they do not directly measure the Sun. They measure how a specific theoretical capture prediction compares with source-calibration data in the gallium detection process.

This immediately implies that an LSC explanation built around BEST can at most claim contact with the source-based gallium anomaly unless it is separately shown to match solar data.

7. Where 27% could really come from

Audit of “27%”: the gallium anomaly depends strongly on the adopted cross section



Source data: Giunti, Li, Ternes et al., Physics Letters B 842 (2023) 137983, Table 2.

Fig. 8. The “27%” value appears naturally in one specific cross-section choice for the gallium anomaly.

The closest rigorous trace of “27%” found during the audit comes from the 2023 inspection by Giunti, Li, Ternes and collaborators. There the global source-anomaly ratio changes significantly with the adopted ^{71}Ga cross-section model. In the Haxton model, the ratio is $R=0.731$, which corresponds to 26.9%. This is an excellent explanation for why 27% may have felt intuitively familiar during project development.

Cross-section model (2023)	Average R	1–R	Interpretation
Ground state only	0.845	15.5%	milder anomaly
Bahcall	0.804	19.6%	close to BEST-like intuition
Haxton	0.731	26.9%	nearest precise literature trace to “27%”
Frekers	0.789	21.1%	BEST-like scale
Kostensalo	0.825	17.5%	reduced anomaly
Semenov	0.787	21.3%	BEST-like scale

Therefore the phrase “27% deficit” can be rehabilitated only in a very controlled sense: it is a model-dependent summary of one gallium-anomaly evaluation, not a universal solar-neutrino fact.

8. The status of the gallium anomaly in 2026

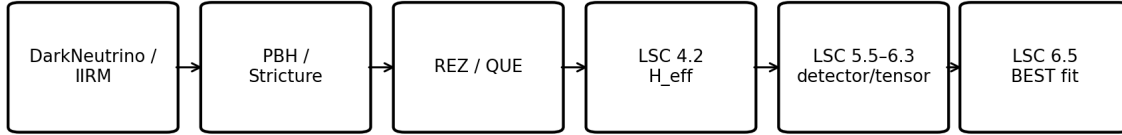
The audit also shows why no rebuilt LSC should lock itself prematurely to a specific anomaly amplitude. In 2026, new self-consistent calculations of the gallium capture cross section moved the preferred ratios upward again, reducing the apparent deficit into the $\approx 16\text{--}19\%$ range for some treatments. An accepted PRL goes even further, arguing that a more refined treatment beyond a simplified leptonic-wavefunction factorization can significantly alleviate or remove the anomaly without invoking new neutrino-sector physics.

That does not mean the anomaly is definitely gone. It means the correct scientific attitude is to let the nuclear-theory uncertainty propagate into the LSC fit. In practice, LSC should scan over reasonable cross-section treatments rather than absorb the entire effect into a new physics parameter by fiat.

PART III

RECONSTRUCTING LSC

Development line of LSC — what to preserve, but with a corrected status



historical inspiration → mechanism hypotheses → effective formalism → data confrontation

The rebuilt monograph does not treat any stage as established physics without a baseline test.

Fig. 9. A corrected developmental line of the project.

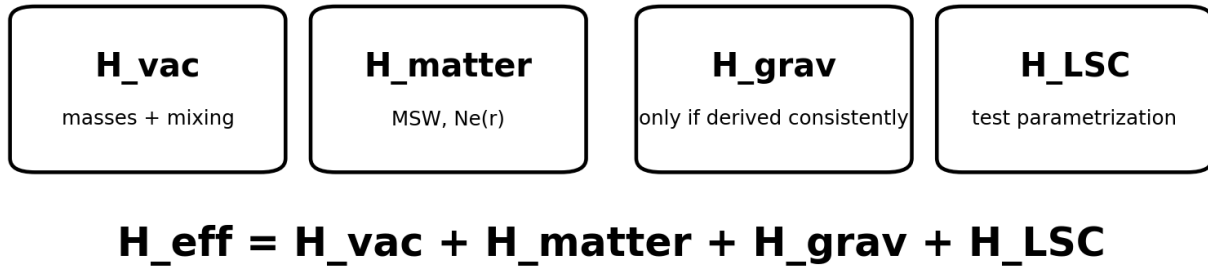
9. Model lineage

The project passed through several conceptual phases: DarkNeutrino / IIRM, PBH / Stricture-zone reasoning, REZ and QUE ideas, the explicit LSC 4.2 Hamiltonian layer, and the later detector/tensor reinterpretations in LSC 5.5–6.3, ending in an LSC 6.5 fit centered on BEST. The rebuilt book does not erase those stages. Instead, it reclassifies them.

- Historical stages are valuable as sources of hypotheses, not as evidence.
- PBH, Stricture, REZ and QUE become candidate mechanisms, not pre-validated explanations.
- The single-neutrino journey becomes a pedagogical illustration, not the statistical foundation of the thesis.
- Every later LSC stage must now be interpreted through a data-first pipeline.

10. Minimal mathematical formalism

Minimal comparative formalism



Null model M0: $H_{\text{LSC}} = 0$. Any new term must improve the data description without violating existing bounds.

Fig. 10. Minimal comparative Hamiltonian structure.

At minimum, the rebuilt framework needs a clean distinction between the null model and the extended model. The standard baseline is

$$H_{\text{std}} = H_{\text{vac}} + H_{\text{matter}}$$

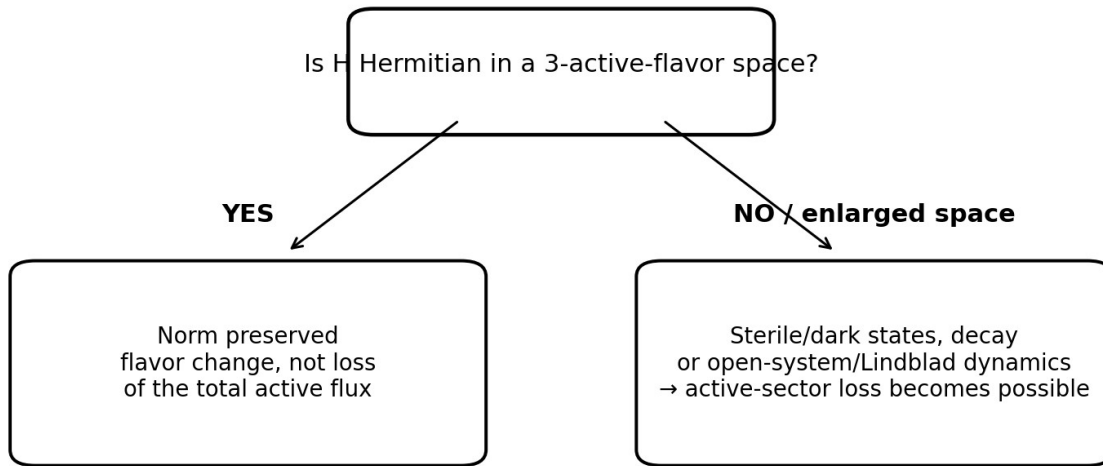
and the test model becomes

$$H_{\text{eff}} = H_{\text{vac}} + H_{\text{matter}} + H_{\text{grav}} + H_{\text{LSC}}$$

The role of H_{grav} and H_{LSC} must be modest at first: they are bookkeeping devices for additional hypotheses. The burden of proof lies on the fit. If the data do not require the extra terms, the proper conclusion is a bound, not an insistence on the mechanism.

11. Unitarity and genuine flux loss

What can “disappearing neutrinos” really mean?



This mathematical choice must come before any story about “27% lost neutrinos”.

Fig. 11. A Hermitian three-flavor Hamiltonian can rearrange flavor, but not destroy the total norm.

This is the deepest mathematical correction to the old story. If the propagation of neutrinos is governed by a Hermitian Hamiltonian in the space of the three active flavors, the evolution is unitary and the norm of the state is preserved. In that case one may suppress ν_e , but one does not reduce the total active population. To obtain a genuine active-flux loss, one must either open the system, include sterile or dark states, or introduce decay-like channels.

$$d\rho/dr = -i[H, \rho] + D_{\text{LSC}}[\rho]$$

This means that any phrase such as “LSC explains 27% missing neutrinos” must first answer a prior question: missing from what space? If the model never leaves the active three-flavor space, then the claim can only refer to flavor redistribution or to detector response, not to a literal disappearance of total active neutrinos.

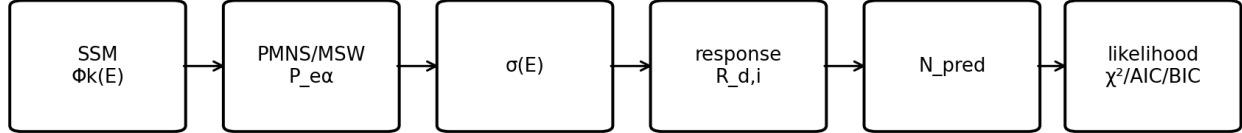
12. PBH, Stricture, REZ and QUE after the status correction

The rebuilt treatment preserves the project’s creative ideas but demotes them to the correct logical level. PBH, Stricture zones, REZ and QUE are now mechanism candidates that must reproduce an already quantified residual. They are no longer allowed to dictate the residual in advance. This is a major conceptual improvement, because it prevents the model from being tuned to a preconceived narrative before the baseline comparison is performed.

PART IV

HOW LSC SHOULD BE TESTED

The pipeline that the old book was missing



Only after reproducing M0 do we add Θ_{LSC} and inspect $\Delta\chi^2$ together with the complexity penalty.

Fig. 12. The analysis chain needed before an LSC claim can be trusted.

13. Data and detector pipeline

The missing piece of the older book was an end-to-end pipeline from solar-model fluxes through oscillations and detector response to a final likelihood. In symbolic form, the predicted event count in detector d and bin i should be written as

$$N_{\text{pred}}(d, i) = T_d \sum_k \int dE \Phi_k(E) \sum_\alpha P_{e\alpha}(E) \sigma_{\alpha d}(E) R_{d,i}(E)$$

Only after this null model is reproduced should an LSC term be turned on. In other words, LSC is not a slogan; it is an extra function inserted into a prediction chain that must already work without it.

Channel	Source	Baseline scale L	Main nuisance	What LSC must do
Solar ν_e , Borexino	Sun	1 AU	solar composition and backgrounds	improve a residual without spoiling $P_{ee}(E)$
Solar 8B, SNO/SK	Sun	1 AU	flavor separation and systematics	respect NC and day-night constraints
Source Ga, BEST	51Cr	~ 1 m	71Ga cross section	distinguish detector/nuclear from propagation effects
Historical Ga solar	Sun	1 AU	integrated response and cross section	stay compatible with solar SNU rates

14. Global fitting and complexity penalties

A proper LSC program compares at least two models:

$$M0 = SSM + PMNS/MSW$$

$$M1 = SSM + PMNS/MSW + LSC$$

The comparison must use a common likelihood or χ^2 and penalize unnecessary complexity through metrics such as AIC, BIC or, if priors are justified, a Bayes factor. The crucial point is that one cannot grant LSC a different free parameter for every experiment. The same physical function must try to explain all relevant data simultaneously.

15. Falsification matrix

LSC falsification matrix

SNO NC	does it change the total active flux?
Borexino	does it preserve $P_{ee}(E)$ across many energies?
Super-K	does it preserve the spectrum and day-night effect?
Solar Ga	does it avoid spoiling long-baseline SNU rates?
BEST	does it explain inner/outer without an artificial extra parameter?
KATRIN/SME	does it avoid independent laboratory limits?

If one parameter set does not pass all channels, the model is not a universal explanation.

Fig. 13. Minimal falsification matrix for a credible LSC program.

Observation	Standard explanation	What LSC might do	What falsifies a simple LSC version
SNO NC total flux	active-flavor conservation under oscillations	predict active leakage or no leakage	significant NC contradiction
Borexino $P_{ee}(E)$	PMNS/MSW energy dependence	add small residual structure	wrong shape across multiple energies
Super-K day-night	Earth-matter effect	introduce extra modulation	failure in zenith/day-night pattern
BEST inner/outer	source-calibration anomaly	fit both with one mechanism	needs ad hoc independent knobs
KATRIN / laboratory bounds	tight limits on exotic effects	remain effectively invisible at low energy	predicts forbidden distortions

16. Computational program and preregistration

A credible next step for the project is to preregister the analysis logic before chasing new mechanisms. That means fixing the datasets, the null model, the cross-section alternatives, the parameter priors, the fit statistics and the decision criteria ahead of time. This protects the program from retrospective narrative drift and from unconsciously tuning the mechanism to match a desired percentage.

PART V

HIGH ENERGIES AND 220 PeV

17. KM3NeT, PBH and why this is a separate branch

The project's later fascination with very-high-energy neutrinos, PBH explanations and 220 PeV events should not be discarded, but it must be placed in the correct branch of the monograph. Solar-neutrino data live in the MeV domain, while KM3NeT and similar astrophysical signals probe energies many orders of magnitude higher. The underlying phenomenology, background structure and relevant constraints are different. Therefore this material belongs to an extension chapter, not to the logical core of the solar-neutrino reconstruction.

In the rebuilt architecture, high-energy neutrinos serve as a stress test for the broader imaginative scope of LSC, especially if the theory proposes geometric or PBH-related modulation mechanisms. But a successful high-energy narrative does not by itself validate a solar-neutrino claim.

PART VI

CONCLUSIONS AND THE NEW PUBLICATION ARCHITECTURE

18. What to keep, rewrite and discard

Element of the old book	Decision	New status
Single-neutrino narrative	keep	move to a later pedagogical chapter
$H_{\text{eff}} = H_{\text{vac}} + H_{\text{matter}} + H_{\text{grav}} + H_{\text{LSC}}$	keep	use as comparative formalism
PBH / Stricture ideas	keep	treat as mechanism hypotheses
REZ / QUE	keep	formalize only after the residual is defined
BEST-centered calibration	keep	explicitly classify as gallium-source, not solar
“27% is an experimental solar fact”	discard	replace with audited, denominator-aware language
“Standard physics does not explain the deficit”	rewrite	only valid after a modern residual analysis
High-energy PBH material as the foundation	relocate	use as a separate extension branch

19. Claims that LSC can really test

After the reconstruction, LSC no longer needs one heroic universal percentage. It needs narrower, sharper claims that can be falsified. Examples include whether an energy-structured residual remains after PMNS/MSW, whether the source-based gallium anomaly survives modern cross-section updates, whether one minimal LSC operator can describe both solar and gallium-source data, whether LSC predicts a distinctive modulation or technology dependence, and whether any PBH/REZ/QUE realization can generate the required amplitude without violating external bounds.

Not: “LSC explains 27% of missing neutrinos.”

But: “LSC defines a class of additional propagation or response modifications whose parameters can be constrained by a joint analysis of solar and source-based neutrino data. A value near 27% appears in one model-dependent variant of the gallium anomaly, but it is not a universal solar fact.”

The new role of the single-neutrino chapter

The old single-neutrino material should be preserved as a final “microscopic walkthrough”: production in the Sun → MSW evolution → optional LSC operator → propagation to Earth → detection. Every step should explicitly mark which part is standard and which is hypothetical. This chapter helps readers understand the equations, but all quantitative statements about percentages and significance must come from populations and likelihoods.

Final conclusion

The greatest gain from the audit is not the discovery of the “right percentage”. It is the removal of the need to possess such a percentage before doing the analysis. If LSC is to contribute something real, it must walk the same road that made neutrino physics successful: precise observables, a baseline, a residual, a minimal extension, a global test and an independent prediction. In the light of the checked project materials, your “27%” should therefore be treated as a working number rather than a foundation. Interestingly, the literature does contain a 26.9% result for the Haxton evaluation of the gallium-source anomaly — a meaningful historical clue, but not a license to call it a solar-neutrino fact.

APPENDIX A

NOTATION AND EQUATIONS

Symbol	Definition
ν_e, ν_μ, ν_τ	active flavor states
$\theta_{12}, \theta_{13}, \theta_{23}$	PMNS mixing angles
Δm^2_{21}	mass-squared difference responsible for the solar scale
$P_{ee}(E)$	electron-neutrino survival probability
$\Phi_k(E)$	flux of component k
$\sigma(E)$	detection cross section
H_{vac}	vacuum Hamiltonian
H_{matter}	MSW matter potential
H_{LSC}	additional operator of the LSC hypothesis
ρ	density matrix
$D_{LSC}[\rho]$	open-system or decoherence operator
$R=N_{obs}/N_{ref}$	observation-to-reference ratio
$D=1-R$	deficit relative to a defined reference

$$V_e = \sqrt{2} G_F N_e$$

$$M0: H_{eff} = H_{vac} + H_{matter}$$

$$M1: H_{eff} = H_{vac} + H_{matter} + H_{LSC}$$

$$D_{active}(E) = 1 - \text{Tr}[P_{active} \rho_{out}] / \text{Tr}[P_{active} \rho_{in}]$$

The last equation represents a genuine active-sector loss only if the dynamics allow transitions outside the three active states or effectively non-unitary evolution.

APPENDIX B

AUDIT TABLES AND QUALITY GATES

B1. Definition Gate for every percentage value

Field	What must be recorded
Experiment	name and phase / run
Dataset	publication version and date
Energy	range or source line
Channel	CC / NC / ES / capture
N_obs	observed value
N_ref	exact reference model
Uncertainties	stat, syst, theory, correlations
D	$1 - N_{\text{obs}}/N_{\text{ref}}$
Status	measurement fact / model output / hypothesis

B2. Reconstruction of the 27% audit

Source	Value	Interpretation
Internal LSC 6.0/6.3	$R_{\text{BEST}} \approx 0.79 \rightarrow \sim 21\%$	gallium source; working model anchor
BEST 2022	$R_{\text{in}} \approx 0.79, R_{\text{out}} \approx 0.77$	$\sim 21\text{--}23\%$ in two zones for one chosen cross section
Giunti et al. 2023, Haxton σ	$R = 0.731$	26.9% — the closest rigorous trace of “27%”
Cadeddu et al. 2026	$R \approx 0.835$ or 0.811	$\sim 16.5\text{--}18.9\%$; strong cross-section dependence
SNO / solar	no single 27%	ν_e suppression and total active flux are different observables

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[P1] LSC 4.2 ULTRA, internal/public project record; Zenodo DOI: 10.5281/zenodo.19602045.

[P2] LSC 6.0, project document and Zenodo record: <https://zenodo.org/records/19780616>. The checked project file used “20% deficit” / $R_{\text{BEST}} \approx 0.79$ rather than a 27% solar deficit.

[P3] LSC 6.3.0 Model Presentation Complete With Archive, internal project material, June 2026.

[P4] LSC Book Complete, previous project book, September 2026; it used BEST inner 0.791 ± 0.05 as an LSC 6.5 calibration point and required a rebuilt narrative axis.

LAST PAGE

One sentence that should guard all future versions

“First define the observation and reproduce the standard baseline. Only then ask whether LSC offers a better explanation of the residual.”

This principle removes the central error of the old book: the model no longer starts from a single neutrino or from a pre-assumed 27%. It starts from the data.